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AUG 03 2016

U. S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387/2016-020-00
UNIT 1 LICENSE NO. NPF-14
PLA-7508**

Docket No. 50-387

Attached is Licensee Event Report (LER) 50-387/2016-020-00. This LER reports a degraded condition concerning a leak in the Reactor Coolant Pressure Boundary. This LER is being submitted pursuant to the requirements of 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(ii)(A) as a condition prohibited by Technical Specifications and as degradation of a principal safety barrier.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new regulatory commitments.


J. A. Franke

Attachment: LER 50-387/2016-020-00

Copy: NRC Region I
Mr. J. E. Greives, NRC Sr. Resident Inspector
Ms. T. E. Hood, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

**LICENSEE EVENT REPORT (LER)**(See Page 2 for required number of
digits/characters for each block)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by internet e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME Susquehanna Steam Electric Station Unit 1	2. DOCKET NUMBER 05000387	3. PAGE 1 of 3
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4. TITLE Reactor Coolant Pressure Boundary Leakage at LPRM Housing as a result of IGSCC

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
06	08	2016	2016	- 020	- 00	08	03	2016	FACILITY NAME	DOCKET NUMBER
										05000
										05000

9. OPERATING MODE 4	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☒ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
10. POWER LEVEL 000	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT M. Krick, Senior Engineer - Nuclear Regulatory Affairs	TELEPHONE NUMBER (Include Area Code) (570) 542-1818
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13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
X	AD	PEN	GO80	Y	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED ☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO	15. EXPECTED SUBMISSION DATE MONTH DAY YEAR
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ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On June 8, 2016 at approximately 02:26, while performing under vessel inspections, a Reactor Coolant Pressure Boundary (RCPB) leak was found at the Local Power Range Monitor (LPRM) 24-09 housing above the housing flange, on the LPRM housing tube. The leakage was determined to be non-isolable from the reactor vessel.

The cause of the RCPB leakage was determined to be Intergranular Stress Corrosion Cracking (IGSCC). Corrective actions to repair the leak have been completed. In addition, a visual inspection was performed on all the Unit 1 LPRM, Intermediate Range Monitor (IRM), and Source Range Monitor (SRM) In-core monitor housings and no further issues were identified.

There was no operational impact as a result of this event due to the plant being in Mode 4 at the time of discovery. This event resulted in an eight (8) hour Emergency Notification System (ENS) communication pursuant to 10 CFR 50.72(b)(3)(ii)(A). This Licensee Event Report (LER) is being communicated pursuant to 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(ii)(A) as a condition prohibited by Technical Specifications and as degradation of a principal safety barrier.

NRC FORM 366A
(11-2015)

U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104

EXPIRES: 10/31/2018



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by internet e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station, Unit 1	05000387	YEAR	SEQUENTIAL NUMBER	REV NO.
		2016	- 020	- 00

NARRATIVE

CONDITIONS PRIOR TO EVENT

Unit 1 - Mode 4, 0 percent Rated Thermal Power

Unit 2- Mode 1, 100 percent Rated Thermal Power

There were no systems, structures or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

On June 8, 2016 at approximately 02:26, while performing under vessel inspections during a plant maintenance outage, a Reactor Coolant Pressure Boundary (RCPB) leak was found at the Local Power Range Monitor (LPRM) 24-09 housing [EIS Code: PEN] above the housing flange, on the LPRM stub housing. The leakage was determined to be about 2 drops per minute (DPM) with the reactor [EIS Code: RPV] de-pressurized and non-isolable from the reactor vessel. Further investigation, including Non-Destructive Examination (NDE) was performed. The through-wall crack was characterized as tight, linear and approximately 1" in length, running parallel with the adjacent weld. The location of the crack was about 0.25 inches from the LPRM housing to flange weld in the heat affected zone (HAZ). This location is also considered not fully mitigated by Hydrogen Water Chemistry (HWC) due to configuration allowing the presence of stagnant water.

Although formal laboratory analysis could not be performed on the crack, the cause of the RCPB leakage was determined to be Intergranular Stress Corrosion Cracking (IGSCC). The LPRM housing is fabricated of SA182- 304 Stainless Steel, which is susceptible to IGSCC. Additionally, this location is subject to reactor pressures, resulting in tensile stresses on the LPRM housing.

There was no operational impact as a result of this event due to the plant being in Mode 4 at the time of discovery. This event resulted in an eight (8) hour Emergency Notification System (ENS) communication pursuant to 10 CFR 50.72(b)(3)(ii)(A). This Licensee Event Report (LER) is being communicated pursuant to 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(ii)(A) as a condition prohibited by Technical Specifications and as degradation of a principal safety barrier.

CAUSE OF EVENT

The cause of the RCPB leakage was the determined to be Intergranular Stress Corrosion Cracking (IGSCC).



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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Susquehanna Steam Electric Station, Unit 1	05000387	2016	- 020	- 00

NARRATIVE

ANALYSIS/SAFETY SIGNIFICANCE

Technical Specification Limiting Condition for Operation (LCO) 3.4.4, RCS Operational Leakage, requires no pressure boundary leakage and less than or equal to 5 gallons per minute (GPM) unidentified leakage when operating in Modes 1, 2 and 3. At the time of discovery, Unit 1 was in Mode 4.

The 5 GPM Technical Specification limit for unidentified leakage is a small fraction of the calculated flow from a critical crack in the primary system piping. Crack behavior from experimental programs show that leakage rates of hundreds of gallon per minute will precede crack instability. As such, the Technical Specification limit allows time for corrective action to be taken before the RCPB could be significantly compromised. The RCPB leak identified during this event was about 2 drops per minute (DPM) with the reactor in depressurized conditions. It is estimated that during rated plant operations, the leakage would have been about 68 DPM or about 0.0009 GPM. Using the crack length identified by Ultrasonic Testing, the point at which the crack became through-wall is estimated to have occurred about 10 months prior to discovery. Based on the information above, including the size of the crack, and the estimated maximum leakage rate, this leak was a small contributor to the actual drywell unidentified leakage of about 0.53 GPM prior to plant shutdown and resulted in no actual consequence to public health or safety.

CORRECTIVE ACTIONS

The corrective action to address this event was to repair the through wall leak on the LPRM 24-09 housing using an ASME Section XI weld repair. In addition, a visual inspection was performed on all the Unit 1 LPRM, IRM, and SRM In-core monitor housings and no further issues were identified. Lastly, an extent of condition review will be performed through visual inspections of the Unit 2 LPRM, IRM and SRM in-core housings during the next refueling outage.

No other corrective actions are required for this event.

COMPONENT FAILURE INFORMATION:

Vendor Component: General Electric In-core Housing

Material: SA182-304

PREVIOUS SIMILAR EVENTS

LER 50-388/ 2015-004-00, PLA-7337, "Degraded Condition Due to Reactor Coolant Pressure Boundary Leakage Caused by Vibration and Stiff Pipe Connection." June 10, 2015.

LER 50-387/ 2015-009-00, PLA-7419, "Pressure Boundary Leakage from an Inadequate Weld Repair in Small Bore Pump Seal Vent Piping." January 11, 2016.

LER 50-387/ 2014-011-00, PLA-7286, "Degraded Condition Due to Reactor Coolant Pressure Boundary Leakage Caused by an Inadequate Weld." February 11, 2015.